^{36}Ar(d,^{3}He) 1974Do12

$J^\pi=0^+$ for ^{36}Ar ground state.

1974Do12: a 52-MeV deuteron beam was produced from the Karlsruhe isochronous cyclotron at the Max-Plank-Institut fur Kernphysik. The target was 200-Torr 99.9% enriched ^{36}Ar gas. ^{3}He particles were detected using detector telescopes consisting of a 200−μm ΔE and a 2-mm E-surface-barrier counter with FWHM=125 keV. Measured $\sigma(E_{^3He},\theta)$. Deduced levels, J, π, L-transfer, and spectroscopic factors from DWUCK-DWBA analysis.

Other: 1975Wa17 (theoretical discussions).

^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{exp}/\sigma(\theta)_{DWBA}/N$, where N=2.95 is a normalization factor used by 1974Do12 from 1966Ba54.

E(level)	J^π‡	L†	C^2S†	E(level)	J^π‡	L†	C^2S†	E(level)	J^π‡	L†	C^2S†
0	3/2$^+$	2	2.2	3170 *20*	7/2$^-$	3	0.22	6750 *20*	(5/2)$^+$	2	0.79
1220 *20*	1/2$^+$	0	1.34	5150 *20*	(5/2)$^+$	2	0.09	6930 *20*	(5/2)$^+$	2	0.46
1763			<0.02	5600 *20*	(5/2)$^+$	2	0.47 *19*	8010 *20*	(5/2)$^+$	2	0.23
2700 *20*	3/2$^+$	2	0.39	5720 *20*	(5/2)$^+$	2	0.73 *29*	8210 *20*	(5/2)$^+$	2	0.28
3000 *20*	5/2$^+$	2	1.49	6140 *20*	(5/2)$^+$	2	0.72	8610 *20*	(5/2)$^+$	2	0.34

† From DWBA analysis of the measured $\sigma(\theta)$ in 1974Do12.

‡ As given in 1974Do12 for levels below 5 MeV. For levels above 5 MeV, 1974Do12 states that they tend to interpret them as 5/2$^+$ states because the summed low-lying $C^2S(1d_{3/2})$ already exceeds the simple shell model value for ^{36}Ar and the measured $\sigma(\theta)$ resembles more closely that of the 5/2$^+$ state at 3.0 MeV than that of the 3/2$^+$ ground state. All C^2S values are given according to the J values.